

# Odinn Waguespack

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## EDUCATION

### University of Maryland

B.S. Mechanical Engineering; GPA: 3.42

College Park, MD

December 2024

### Scholars Honors College

Science, Technology, and Society Program

College Park, MD

Citation Conferred May 2022

## EXPERIENCE

### Carnegie Robotics

Mechanical Engineering Intern

Pittsburgh, PA

May 2025 - Sep 2025

- Assisted in the ground-up development of a fully self contained off-road autonomy platform
- Designed and installed a LiDAR, Stereo Camera and IMU perception payload on a Polaris RZR. Modeled vehicle kinematics to inform FOV geometry
- Redesigned shift-linkage and electronic power steering mounting for conversion to drive-by wire autonomy
- Performed thermal/flow-sim analysis of active ventilation, resulting in a 11% decrease to allocated power budget

### National Institute Of Standards And Technology

Mechanical Engineering Intern - Quantum Measurement Division

Gaithersburg, MD

Oct 2024 - April 2025

- Developed a high precision torsion pendulum for integration with an atom interferometer testing the quantum coherence of gravity
- Implemented python scripts to automate LabView data collection
- Oversaw CAD design and manufacture of precision mechanics, optics, and vacuum components
- Designed an experimental ultra-high-vacuum (UHV) test setup and managed part requisition, decreasing project lead-time by 4 months

### Terrapin Works - Machine Shop

Technical Supervisor

College Park, MD

Dec 2022 - Dec 2024

- Managed all 5-axis CNC operations and machine maintenance
- Oversaw training of new machinists on CNC, Waterjet, and Lathe operations

### Agentis Air

Mechanical Engineering Intern

Derwood, MD

May 2023 - August 2023

- Researched a novel ionizing filtration system. Integrated prototype with existing commercial product, increasing system efficiency by 24%
- Wrote C++ scripts for real-time monitoring of Optical Particle Counter (OPC) sensor data, reducing individual test times by 30 minutes

### University of Maryland

Teaching Assistant - CPSP369 Robotics Service Learning

College Park, MD

Aug 2022 - Dec 2022

- Hosted robotics outreach lab for students grades 6-8
- Received highest positive course rating (5/5) from student feedback

## PROJECTS AND ACTIVITIES

### Robotic Manipulator, Senior Capstone (1st Place)

Mechanical Engineer

College Park, MD

Jan 2024 - May 2024

- Worked with a team of five to design a 3-DOF robotic manipulator arm for integration with the Blue-ROV2 platform
- Developed an SLA printed actuator for manipulator control, focusing on optimizing applied force and IP68 waterproof rating
- Improved project cohesion by organizing design reviews to ensure technical specifications and deadlines were met
- Placed 1st out of 37 competing projects

### Formula SAE

Power Train Engineer

College Park, MD

Aug 2021 - Dec 2024

- Designed a 4-1 exhaust manifold for optimization of scavenging, resulting in a 12% increase in peak torque at 10k rpm
- Designed and CNC milled custom oil tank mounts

### MiniBot V2

Personal Project

College Park, MD

Aug 2025 - Present

- Built a 5.4lb, 5-DOF bipedal robot capable of semi-autonomous terrain traversal
- Redesigned electronics and compute mounting solutions for lowered COG, resulting in increased locomotion stability
- Currently training improved walking policy using MuJoCo-based physics simulation environment

## SKILLS AND AWARDS

Awards: Capstone Design Expo Deans Award, Deans List Fall 20-21, Spring 23-24

Programming: Python, C++, Matlab, MuJoCo, ROS2

Design/Software: Solidworks, Autodesk Fusion 360, Siemens NX, Excel, FEA, ANSYS, Creo, Overleaf/Latex

Manufacturing: Multi-axis CNC, Lathe, Manual Machining, 3D Printing (SLS, SLA, FDM, Powderbed), Laser Cutting, Waterjet